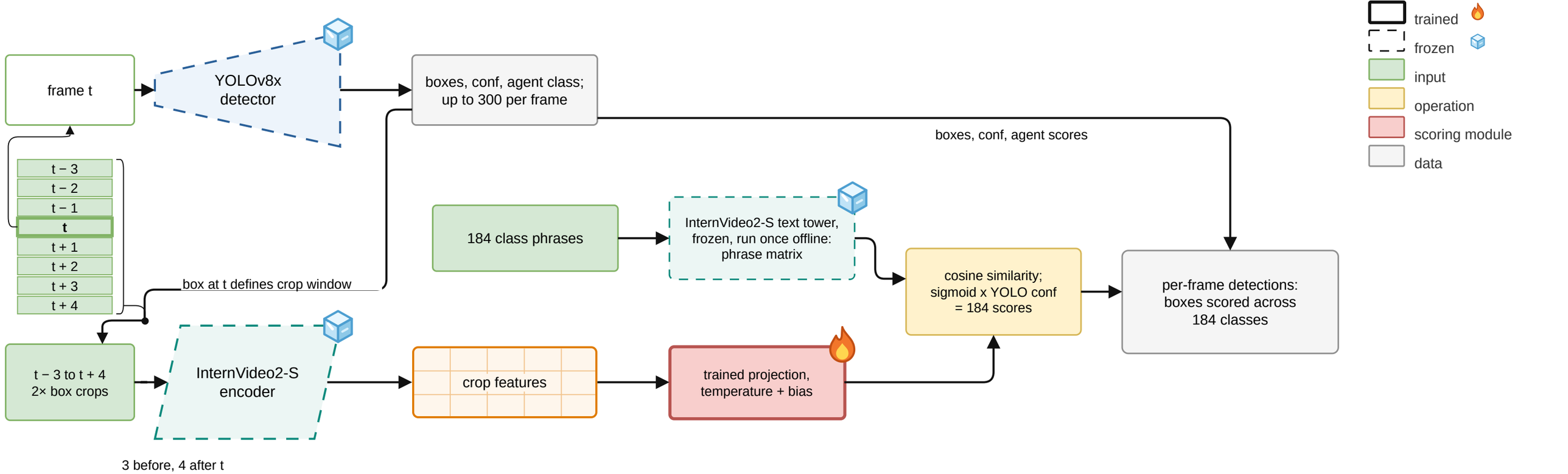

frame t
YOLOv8x detector
boxes, conf, agent class; up to 300 per frame
boxes, conf, agent scores
t − 3
t − 2
t − 1
t
t + 1
t + 2
t + 3
t + 4
box at t defines crop window
184 class phrases
InternVideo2-S text tower, frozen, run once offline: phrase matrix
cosine similarity; sigmoid x YOLO conf = 184 scores
per-frame detections: boxes scored across 184 classes
t − 3 to t + 4 2× box crops
InternVideo2-S encoder
crop features
trained projection, temperature + bias
3 before, 4 after t
trained
frozen
input
operation
scoring module
data